IDX G9 BIOLOGY H STUDY GUIDE ISSUE S2 FINALS

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This guide is a combined guide. Due to the final range including content from before the monthlies, we have included content from the previous issue in this guide for your convenience.

18.1 Finding Order in Diversity

- **Binomial Nomenclature**
  - 1730s, Swedish, Carolus Linnaeus
  - Two - part scientific name in italics (e.g., polar bear: Ursus maritimus)
    - First word (capital) - Genus
    - Second word (lower) - Species.
    - Underlined if written by hand
- **Dichotomous key**: Used to identify organisms.
  - Series of paired statements/questions that describe possible characteristics of the organism (Yes/No)
- **Traditional classification**: Based on morphology (form & structure)
- **Systematics**: Science of naming and grouping organisms
  - Goal: Organize living things into groups (taxa) that have biological meaning.

    - Assign each species a unique, universally accepted name
- **Linnaean Classification System** (7 hierarchical taxa): species → Genus → Family → Order → Class → Phylum → Kingdom
- Modern systematics: Looks beyond simple similarities/differences; asks questions about evolutionary relationships

18.2 Modern Evolutionary Classification

- **Evolution**: Descent with modification
- **Phylogeny**: Study of evolutionary history
    - Phylogenetic Systematics: evolutionary classification, to group species into categories that reflect lines of evolutionary descent, rather than overall similarities and differences
- **Cladistics**: Method of classifying species of organisms into groups (**clades**), which consist of an ancestor organism and all its descendants (nothing else)
    - Clades must be monophyletic group — includes all species descended from a common ancestor
- **Cladistic analysis**: shows many taxonomic groups from clades
    - e.g. mammalia
    - Valid clades example: clade reptilia includes birds (class doesn't). Crocodiles more closely related to birds
- **Cladograms**: Tree diagram to link clades together
    - Links by showing how evolutionary lines (lineages) branch off from common ancestors
    - Bottom (root) of cladogram represents common ancestor shared
    - Branching points: **nodes**
        - Nodes denote a speciation event when common ancestor split into two or more species

    - Branching patterns indicate degrees of relatedness among organisms
- **Derived character**: Trait that arose in the most recent common ancestor of a particular lineage and was passed along its descendants
    - Four limbs: clade tetrapoda, but not derived for mammals
    - Hair: clade mammalia
    - Specialized shearing teeth: clade carnivora
    - Retractable claws: clade Felidae (cats)
- Absence of trait ≠ derived character, because distantly related organisms can lose same character.
    - e.g., both whales & snakes lost tetrapod character of 4 limbs, but whales - clade mammalia, snakes - clade Reptilia
- **Genetics**: Primary basis for grouping organisms to clades & determining paths of evolutionary descent
    - DNA tools make it more accurate
- Because genes mutate, shared genes have differences = derived characteristics in cladistic analysis
    - Similarities & differences in DNA build hypotheses about evolutionary relationships.
        - e.g., llamas put in camel family (originally sheep)

18.3 Building the Tree of Life

Changing Ideas About Kingdoms

- 1700s: Linnaeus' original system divided all organisms into plants (**Plantae**) and animals (**Animalia**)
- Late 1800s: **Protista** was added for microorganisms (protists and prokaryotes)
- 1950s: **Monera** was added for prokaryotes; **Fungi** was added

    - **Prokaryotes**: no membrane-bound organelles, no nucleus
    - **Eukaryotes**: membrane-bound organelles, nucleus
- 1990s: Monera was split into **Eubacteria** and **Archaebacteria**
- Genomic analysis revealed that eukaryotes and the two prokaryotic groups are more different from each other than previously thought.
    - New taxonomic category—**Domain**: larger, more inclusive category than a kingdom
- Kingdom "Protista": paraphyletic group (not a true clade)
    - Only monophyletic groups are valid under evolutionary classification

The Tree of Life

- Represents evolutionary relationships among the three domains.
- Domain **Bacteria** (Kingdom **Eubacteria**)
    - Unicellular and prokaryotic, autotrophic/heterotrophic, cell walls with peptidoglycan
    - Ecologically diverse, can be aerobic/anaerobic
        - Aerobic: requiring oxygen to survive
        - Anaerobic: can only survive in the absence of oxygen
    - E.g., E. coli, cyanobacteria
- Domain **Archaea** (Kingdom **Archaebacteria)**
    - Unicellular and prokaryotic, autotrophic/heterotrophic, cell walls lack peptidoglycan, live in some extreme environments
    - Many are anaerobic
    - E.g., halophiles
- Domain **Eukarya**: eukaryotic
    - Cells have nucleus (that contains DNA)
    - Kingdom **Protista**
        - Mostly unicellular, autotrophic/heterotrophic, some have cell walls with cellulose

        - Some display characters that resemble those of fungi, plants or animals
        - Not a true clade
        - E.g., amoeba, brown algae
    - Kingdom **Fungi**
        - Mostly multicellular, heterotrophic, cell walls with chitin
        - Most feed on dead/decaying organic matter by secreting digestive enzymes
        - E.g., mushrooms, yeasts
    - Kingdom **Plantae**
        - Multicellular, autotrophic, cell walls with cellulose
        - Non-motile
        - E.g., mosses, ferns, flowering plants
        - (The textbook says green algae are unicellular plants, but theories are changing)
    - Kingdom **Animalia**
        - Multicellular, heterotrophic, no cell walls
        - Mostly motile
        - E.g., sponges, worms, mammals

19.1 The Fossil Record

- Fossil: the preserved remains / traces of ancient life
- From fossil record, paleontologists learn about the structure of ancient organisms, their environment, the ways in which they lived
    - tiny percentage become fossils under special conditions
- Fossils found in rock include petrified fossils, molds & casts, trace fossils

- Petrified fossils →minerals replace all/part of an organism

- Trace fossils evidence of activities of ancient organisms
    - Ex: footprint, burrow
- Mold: hollow area in sediment in the shape of an organism
- Cast: copy of shape of an organism

- Other fossils form when remains of organisms are preserved in substances such as tar, amber, ice
- Amber fossils: become trapped in tree resin that hardens after the tree is buried

- Fossils usually found in sedimentary rock
- Sedimentary rock forms when small particles of sand, silt, clay, lime muds settle down to bottom of body of water
- sediment build up, bury dead organisms
- Over many years, water pressure gradually compresses

- Relative dating places rock layers and their fossils into a temporal sequence determine a fossil younger/older
    - Index fossils are used to establish and compare the relative ages of rock layers and the fossils they contain
        - existed only in a few rock layers
        - occurred in large geographic areas
        - Index fossils include trilobites
- Radiometric dating relies on radioactive isotopes, which decay into nonradioactive isotopes at a steady rate
    - compares the amount of radioactive to nonradioactive isotopes in a sample

  - half life is the time required for half of the radioactive atoms in a sample to decay
  - Carbon-14 has a half life of 5730 years, only useful for dating fossils no older than 60,000 years
  - most common: potassium-40 and uranium-238

- Timeline of Earth's history : geologic time scale
- basic divisions: eons, eras, periods
- geologic time before Cambrian is Precambrian time (90% of history)
- Eras consists 2 or more periods → Cenozoic, Mesozoic, Paleozoic
- Periods associated with rock systems
- Physical forces: The theory of plate tectonics explain how continental plates move slowly above earth's molten core → continental drift.

19.2 Patterns and processes of evolution

Macroevolution vs Microevolution

- Microevolution: change in allele frequencies that occurs overtime within a population
- Macroevolution: grand transformations in anatomy, phylogeny, ecology, behavior

Macroevolution & cladistics

- Fossils are classified through shared derived characters
- Fossils are placed in clades that contain only extinct organisms/ include living organisms

- If rate of speciation in a clade is equal to/greater than the rate of extinction, the clade will continue to exist
- If rate of extinction >rate of speciation , clade extinct

Patterns of Extinction

① Background extinction: extinct because of slow steady process of natural selection

② Mass extinction: relatively short period of time in which entire ecosystems vanish, whole food web collapse

- natural selection can't compensate quickly
- Ex: volcanic eruptions, moving continents, changing sea level
- End of cretaceous period: asteroid crushed into earth

Pace of Speciation

① Gradualism

- Darwin
- Evolution occurred by a long sequence of continuous change
- Ex: the change in size and hoof of modern horse

② Punctuated equilibrium

- long period of relative stability in a species are punctuated by periods of rapid evolution
- no long sequence of intermediate forms
- Ex: horseshoe crab
- much more common in organisms with short generation time

Patterns of Evolution: adaptive radiation, convergent evolution, coevolution

- Coevolution: Two Species evolve in response to changes in each other over
- ①Flower & Pollinators →orchids with long spur and moths with long feeding tube
- ②Plants &herbivorous insects → plant evolved bad tasting/poisonous compound; Insects evolved variants that could alter, inactivate, eliminate poison

19.3 Earth's Early History

- Formation of Earth
    - cosmic debris collided with one another
- Earth 4.6 billion years ago
    - violent volcano eruptions
    - comets and asteroids bombarded
- Earth 4.2 billion years ago
    - solid rocks
    - water condensed, rain
    - Ocean, with lots of dissolved iron
    - atmosphere little/no oxygen, principally carbon dioxide, water vapor, nitrogen, with lesser amounts of carbon monoxide, hydrogen sulfide, hydrogen cyanide

First organic molecules

- 1953 Stanley Miller & Harold Urey experiment suggested how mixture of organic compounds could have arisen from simpler compounds

Formation of Microsphere

- During Archean Eon
- Large organic molecules form tiny bubbles called proteinoid microsphere
- microspheres
    - have selectively permeable membrane for water to pass
    - have simple ways of storing and releasing energy

Evolution of RNA and DNA

- RNA hypothesis
    - RNA formed from simpler molecules and existed before DNA
    - Simple RNA could lead to DNA directed protein synthesis.

Production of free oxygen

- prokaryotes -first life form→3.5 billion years ago, absence of oxygen
- photosynthetic bacteria producing oxygen 2.5 billion years ago
    - Oxygen combined with iron in ocean
    - Oxygen accumulates in atmosphere, forming ozone layer